

Juana Granados Rodríguez

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SUMMARY

Theoretical physicist with an MSc in Astrophysics, Particle Physics and Cosmology, specializing in Particles and Gravitation, and a BSc in Physics, with research experience spanning quantum field theory, mathematical physics, renormalization-group methods, statistical field theory, and condensed-matter physics. My research has included mathematical analysis of quantum-field-theoretic distributions and propagators, symmetry breaking and the hierarchy problem using lattice/Ising-type models and RG flows, and experimental nanomagnetism. I am interested in doctoral research in theoretical physics, statistical and non-equilibrium physics, complex systems, and applied mathematics, particularly where field-theoretic and mathematical methods can be applied to collective phenomena.

EDUCATION

- Sept 2025 – Jul 2026 **MSc in Astrophysics, Particle Physics and Cosmology**
Specialization: Particles and Gravitation
Universitat de Barcelona, Spain
Selected Coursework: Quantum Field Theory (9.9/10), Elementary Particles (9.0/10), Gauge Theories of the Standard Model (8.0/10), Mathematical and Statistical Techniques (9.3/10)
Final grade: 9.24/10
- Jan 2020 – Jul 2024 **BSc in Physics**
Universidad de los Andes, Bogotá, Colombia
Selected Coursework: Particle Physics (4.20/5.00), Perturbative Renormalization (4.72/5.00)
Recipient of the *Préstamo Condonable para Carreras Especiales* and *fully condoned for academic merit*.
Final grade: 4.28/5.00
- Minor in Mathematics**
Universidad de los Andes
Coursework: Advanced Linear Algebra, Abstract Algebra, Complex Calculus, Probability (Honors)

RESEARCH EXPERIENCE

- Master's Thesis, Universitat de Barcelona** Feb 2026 – Jul 2026
[Symmetry Protection and the Hierarchy Problem: From Lattice Models to Goofy Transformations](#)
Matrícula d'Honor (Distinction), 9.8/10
Supervisor: [Associate Professor Jordi Salvado Serra, Ph.D.](#)
Investigated symmetry protection and the hierarchy problem through complementary lattice and continuum approaches.
Extended an inverse-Ising renormalization-group pipeline to $\mathbb{Z}_2$ -odd operators, modifying Monte Carlo sampling and pseudo-likelihood inference to study their behavior under Kadanoff blocking.
Performed explicit one-loop quantum field theory calculations of fermion and scalar self-energies using dimensional regularization to analyze radiative mass corrections and symmetry protection.
Examined recently proposed GOOFy transformations as a candidate mechanism for protecting scalar masses.
- Research Assistant, Condensed-Matter Physics, [Nanomagnetism Group](#), Universidad de los Andes** Jun 2024 – Jun 2026
Supervisor: [Associate Professor Juan Gabriel Ramírez, Ph.D.](#)
Synthesized and characterized Co-doped V_2O_5 nanoparticles using Vibrating Sample Magnetometry (VSM), X-ray Diffraction (XRD), and Raman spectroscopy.
Performed statistical analysis and interpretation of experimental data using Python (NumPy, SciPy, Matplotlib, Pandas, Scikit-learn), R, and Jupyter Notebooks.
Contributed to the analysis and preparation of a peer-reviewed scientific manuscript and presented research findings in group seminars.
Awarded first place in the experimental category of the university's student research showcase (2024).
Poster: "*Antiferromagnetic Superexchange and Oxygen Vacancies in Co-Doped V_2O_5 Nanoparticles: A DFT+U and Experimental Study*", IN²UB 2026 Annual Inhouse Meeting, Universitat de Barcelona (July 2026).

Bachelor’s Thesis, Universidad de los Andes
The wavefront set and its applications in quantum field theory
Excellent, 4.90/5.00
Supervisor: Associate Professor Andres Fernando Reyes Lega, Ph.D.
Analyzed wavefront sets and Hörmander’s criterion to characterize the singular structure of distributions and determine the well-definedness of their products in quantum field theory.
Studied the wavefront sets of the Wightman and Feynman propagators, including the singularities that obstruct the definition of higher-order propagator products.
Examined Epstein–Glaser renormalization, including scaling degree and the extension of singular distributions in the construction of time-ordered products.

Jan 2024 – Jul 2024

QFT and Mathematical Physics Seminar, Universidad de los Andes
Participated in an advanced seminar on quantum field theory and mathematical physics, initially as a regular participant and later through closer involvement with the research group.
Studied the mathematical structure of wavefront sets and their role in quantum field theory.
Applied Epstein–Glaser techniques to problems involving renormalization and distributions.
Participated in weekly technical discussions and research seminars.
The seminar experience led to the development of the Bachelor’s thesis within the same research group under the supervision of Associate Professor Andres Fernando Reyes Lega.

Jan 2023 – Jul 2024

TEACHING EXPERIENCE & COMMUNITY ENGAGEMENT

Teaching Assistant, Universidad de los Andes
Vector Calculus
Honors Integral Calculus and Differential Equations

Jun – Aug 2023
Aug – Dec 2021

Science Content Creator, Universidad de los Andes
Created and appeared in short-form videos for the university’s official TikTok channel, making science engaging and accessible to young audiences in Colombia.

Jun – Aug 2023

TECHNICAL & RESEARCH SKILLS

Programming & Scientific Computing: Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), R, Jupyter Notebooks
Computational & Statistical Methods: Monte Carlo methods, Markov chains, pseudo-likelihood inference, statistical modelling, scientific data analysis
Theoretical & Mathematical Methods: Quantum field theory, renormalization, renormalization-group methods, distribution theory, wavefront sets, Epstein–Glaser renormalization, lattice/Ising models
Selected GitHub Work: Public project on Ising models and Markov chains; repositories from Computational Methods coursework; master’s thesis simulations.

LANGUAGES

Spanish (Native), English (IELTS Academic: 7.5, C1)